

Robotics & Automation

BRANCH: MECHANICAL ENGG. (ME)

COURSE CODE: ME405

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Robotics & Automation (ME405) under the Mechanical Engg. (ME) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Introduction & Kinematics

Robot definition, laws of robotics, anatomy, joints, workspace. Homogeneous transformations. Forward and Inverse Kinematics: Denavit-Hartenberg (D-H) representation.

Unit 2: Robot Dynamics & Control

Lagrange-Euler and Newton-Euler dynamic formulations. Robot control: Joint control, PID control, feedback control systems.

Unit 3: Sensors & End Effectors

Sensors: Touch, force, proximity, ultrasonic sensors. Machine Vision: image processing, camera calibration. End effectors: mechanical, vacuum, magnetic grippers.

Unit 4: Robot Programming & Industrial App

Lead-through programming, textual programming languages. Industrial applications: Welding, painting, assembly, palletizing, warehouse logistics.

Degrees of freedom of a spatial manipulator. D-H parameter table: theta, d, a, alpha.

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



